AI-DRIVEN SMART CINNAMON GROWTH MONITORING & AUTOMATED HARVESTING SYSTEM

***Abstract*—Ceylon cinnamon (*Cinnamomum verum*) is still harvested through a sequence of decisions and manual operations that depend heavily on experienced workers. A stem must be mature enough to peel, its health and bark condition must be acceptable, and the harvesting tool must interact with the stem without damaging the bark or wood. This paper presents an integrated smart-cinnamon platform that connects multi-environment IoT monitoring, machine-learning-based growth and harvest-readiness assessment, image-based bark analysis, and safety-gated robotic harvesting. Temperature, relative humidity, soil moisture, light exposure, RGB images, and stem morphology are aligned using persistent plant or stem identifiers. The learning layer is designed to compare sensor-only, vision-only, morphology-only, and multimodal models rather than assuming that data fusion is automatically beneficial. A calibrated vision stage identifies the usable stem/bark region and converts image detections into a physical target for the robot. The robot is allowed to act only when readiness, localisation confidence, and workspace checks satisfy predefined limits. The contribution is therefore a decision-to-action architecture for cinnamon in which biological readiness and geometric confidence jointly control harvesting. A proof-of-concept software pipeline has been assembled, while final plant-wise predictive validation and statistically powered physical harvesting trials remain part of the ongoing evaluation. The paper defines a reproducible protocol for measuring readiness prediction, bark-region localisation, coordinate error, peel success, damage, cycle time, and safety aborts without presenting unverified proposal values as experimental results.**

***Index Terms*—Ceylon cinnamon, smart agriculture, Internet of Things, multimodal machine learning, harvest readiness, computer vision, agricultural robotics, robotic harvesting**

I. Introduction

Ceylon cinnamon (*Cinnamomum verum* J. Presl) is a perennial spice crop closely associated with Sri Lankan agriculture. Its value depends not only on growing a healthy plant but also on harvesting stems at an appropriate stage and removing the bark carefully. Studies on Ceylon cinnamon have shown that maturity influences bark-oil quantity and quality, while stick properties such as circumference and weight are related to bark yield [1]–[3]. These observations make harvest readiness a biological decision rather than a simple image-classification task.

Cinnamon processing also contains a difficult physical step: bark peeling. Mechanisation research has demonstrated that vibration assistance and purpose-built peeling mechanisms can reduce manual effort or improve peeling action [4], [5]. However, a peeling mechanism alone does not determine whether a particular stem should be harvested or where a robot should make contact. A useful autonomous system must connect plant-state information to perception and controlled manipulation.

The same integration problem appears in broader agricultural robotics. Modern field robots combine sensing, perception, localisation, motion planning, and task-specific end effectors; recent reviews emphasise that reliability depends on the complete chain rather than on vision accuracy in isolation [6], [7]. Multimodal crop models also show that image and sensor streams can complement each other when estimating plant state [8]. These developments motivate a cinnamon-specific architecture that brings environmental monitoring, machine learning, computer vision, and robotic control into one traceable workflow.

This research addresses the following question: *Can environmental history, visible stem information, and simple morphology be combined to estimate cinnamon harvest readiness, and can that estimate be used safely to gate a calibrated robotic bark-harvesting action?*

The main contributions are:

1) an IoT data model for monitoring cinnamon under multiple environmental conditions while maintaining plant-level traceability;
2) a multimodal machine-learning framework for growth state, bark-health assessment, and harvest-readiness prediction;
3) a vision-to-robot interface that converts valid bark detections into calibrated robot targets instead of using raw image coordinates as physical coordinates; and
4) an end-to-end evaluation protocol that separates biological decision errors, visual localisation errors, robot-control failures, and contact/peeling failures.

II. Related Work and Research Gap

A. Cinnamon maturity and harvesting

Wijeweera *et al.* examined maturity-dependent changes in Ceylon cinnamon and reported that the maturity stage affects the quantity and chemical composition of bark and leaf oils [1]. Ranasinghe *et al.* linked stick characteristics with bark yield, while Aluthgamage *et al.* considered harvestable stem number, harvestable length, and unit bark weight as useful yield indicators [2], [3]. Together, these studies support the inclusion of stem morphology and maturity evidence in a readiness model. They also show why a label such as "ready" should not be generated from a single photograph alone.

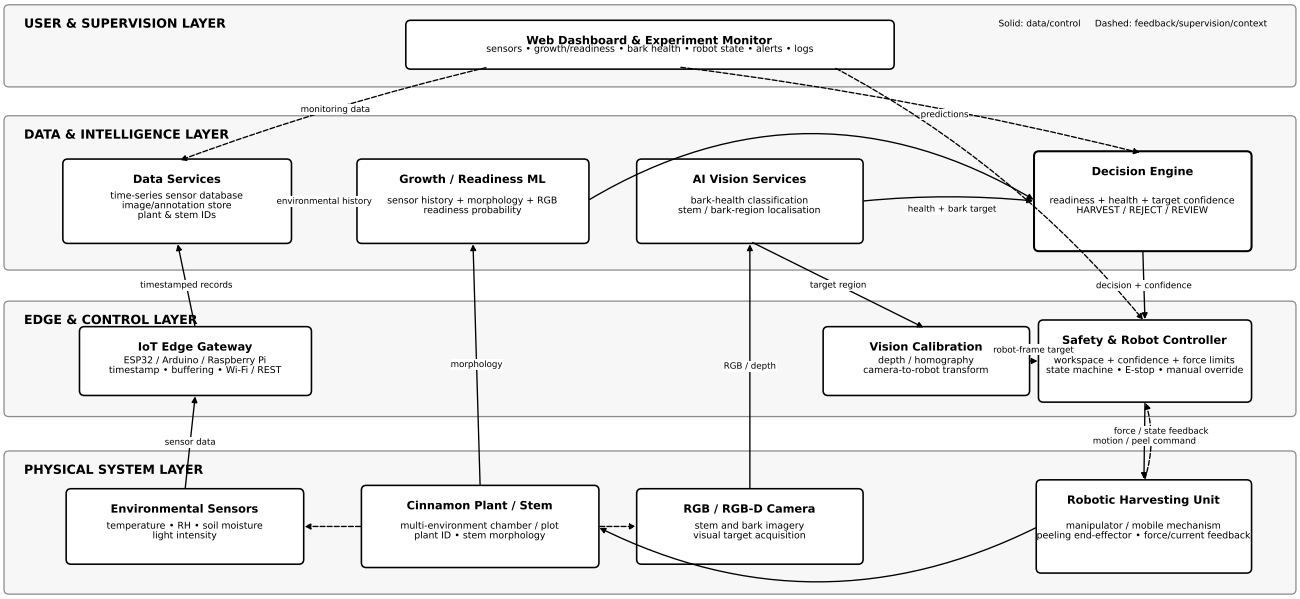


Fig. 1. Layered system architecture of the proposed AI-driven smart cinnamon monitoring and automated harvesting platform. The physical layer contains the cinnamon environment, sensors, camera, and robotic harvester; the edge/control layer handles acquisition, calibration, safety, and robot control; the data/intelligence layer stores synchronized records and executes the ML and vision models; and the supervision layer exposes the dashboard and operator override. Dashed paths denote feedback, logging, or supervisory control.

B. Cinnamon peeling and agricultural robotics

Bandara *et al.* investigated vibration-assisted peeling, and Gunawardhana *et al.* developed a single-stroke 360-degree arc-length mechanism for cinnamon processing [4], [5]. These works focus on the mechanics of peeling, which is complementary to the present research. The missing link is a perception-and-decision layer that decides when robotic peeling is appropriate and where the contact region is located.

In other crops, robotic harvesting has advanced through RGB/RGB-D perception, target localisation, manipulator design, and closed-loop control. Rajendran *et al.* review perception, robot design, motion planning, and control as tightly coupled problems in selective harvesting [6]. Liu *et al.* similarly identify perception, navigation, manipulation, and intelligent decision-making as recurring challenges for field robots [7]. An integrated apple-harvesting system evaluated by Zhang *et al.* further demonstrates the need to measure complete robot performance rather than report detection accuracy alone [9].

C. Multimodal crop modelling

Li *et al.* introduced ViST, a vision-and-sensor transformer that combines RGB imagery with sensor measurements for crop-growth prediction [8]. The key lesson for cinnamon is not that a transformer must be used, but that environmental history and visual evidence may carry complementary information. The present work therefore treats fusion as a testable hypothesis and retains strong single-modality baselines.

D. Research gap

Current cinnamon studies provide agronomic maturity indicators and specialised peeling mechanisms, while smart-agriculture studies provide multimodal prediction and vision-guided robotic methods. What remains underexplored is a cinnamon-specific decision-to-action pipeline in which monitored growth conditions and stem observations determine readiness, vision identifies a physically valid bark region, and robot execution is enabled only after confidence and safety checks. This integration forms the core research gap addressed by the proposed system.

III. System Design and Methodology

Figure 1 presents the complete layered system architecture. Unlike a process-flow diagram, it identifies the deployed physical devices, edge/communication components, data and intelligence services, safety and robot-control components, and the user supervision interface, together with the data, command, feedback, and override connections among them. This separation keeps the IoT, ML, vision, and robotic modules independently testable while preserving an explicit end-to-end interface contract.

A. IoT multi-environment monitoring

The monitoring layer records temperature, relative humidity, soil moisture, and light intensity. Sensors may be deployed in controlled growth chambers, greenhouse-like test spaces, or field plots. Every reading is stored with a timestamp, environment identifier, and persistent plant/stem identifier. This is important because repeated measurements from one plant are

correlated; treating them as independent samples would inflate apparent model performance.

Rather than using only the latest sensor value, the learning pipeline derives time-window statistics. For a sensor variable x observed over window W, the mean is

$$\mu_W = \frac{1}{N}\sum_{i=1}^{N} x_i, \tag{1}$$

and variability is represented by

$$\sigma_W = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \mu_W)^2}. \tag{2}$$

Minimum, maximum, range, and slope can be added for windows such as 24 h, 72 h, and 7 d. Missing values are flagged explicitly, and scaling parameters are estimated from training data only.

B. Plant images, morphology, and health

RGB images capture stem appearance and bark condition. Where available, plant height, stem circumference/diameter, age/history, and a measured or expert-assessed peelability indicator are linked to the same identifier. Image augmentation is restricted to the training partition and uses modest geometric and illumination changes that preserve the biological meaning of the label.

The visual health component is treated as a secondary classifier that can identify labelled bark-health categories, including healthy and disease/insect-damage classes represented in the collected dataset. It should not be presented as a general cinnamon-disease diagnostic model unless the training data cover those conditions with independent ground truth.

C. Harvest-readiness learning

Four model families are compared: environmental-only, morphology-only, RGB-only, and multimodal fusion. Classical tabular baselines such as logistic regression, Random Forest, and gradient boosting provide interpretable comparisons for sensor and morphology features. A compact transfer-learning CNN or similar visual encoder produces image features $\mathbf{z}_v$. Encoded sensor and morphology features form $\mathbf{z}_s$. A simple late-fusion baseline is

$$\mathbf{z}_f = [\mathbf{z}_v; \mathbf{z}_s], \tag{3}$$

with readiness probability

$$\hat{y} = \sigma\left(W_2\phi(W_1\mathbf{z}_f + b_1) + b_2\right). \tag{4}$$

This architecture is intentionally simple enough to determine whether fusion itself adds value before more complex cross-attention models are considered.

TABLE I
MAIN SYSTEM MODULES AND RESEARCH OUTPUTS

Module	Main inputs	Main outputs
IoT monitoring	Temperature, RH, soil moisture, light	Time-aligned environmental history
Growth/readiness ML	Sensor windows, morphology, RGB features	Growth state and readiness probability
Bark-health vision	RGB stem/bark image	Health class and confidence
Harvest vision	RGB/RGB-D image, calibration	Bark region and robot-frame target
Robot controller	Readiness, target, safety state	Approach/peel/abort action and trial log

D. Vision-guided target generation

A central practical issue is coordinate validity. A 2-D image coordinate (u, v) cannot be sent directly to a three-dimensional robot as an (x, y, z) position. With a calibrated camera matrix K and depth z, an image point $\mathbf{p} = [u, v, 1]^T$ can be reconstructed in the camera frame as

$$\mathbf{x}_c = zK^{-1}\mathbf{p}. \tag{5}$$

A calibrated camera-to-robot transform T_{rc} then gives

$$\begin{bmatrix}\mathbf{x}_r \\ 1\end{bmatrix} = T_{rc}\begin{bmatrix}\mathbf{x}_c \\ 1\end{bmatrix}. \tag{6}$$

If depth sensing is unavailable, a constrained laboratory setup can instead use a fixed camera, known working plane, reference marker, and validated homography. The implemented geometry must be reported explicitly so that coordinate accuracy can be measured in millimetres.

E. Safety-gated robotic harvesting

The controller receives both a biological decision and a geometric confidence. Let P_r be readiness probability and C_v the localisation confidence. A harvest command is accepted only if

$$P_r \geq \tau_r, \qquad C_v \geq \tau_v, \tag{7}$$

and the target lies inside a validated workspace $\mathcal{W}_{safe}$. The robot first moves to a pre-contact pose, approaches at reduced speed, and starts the peeling action only after the target is revalidated. A force sensor, load cell, or calibrated motor-current proxy can provide an additional stop condition. Emergency stop, timeout, motion limits, and manual override are required during physical testing.

IV. Dataset and Experimental Protocol

A. Ground truth and data partitioning

A strong readiness label must be independent of the model's visual input. The recommended protocol combines expert agricultural judgement with measurable evidence such as plant age/history, stem circumference, bark appearance, and a standardised peelability assessment. Because cinnamon is perennial, a short student project should distinguish between a longitudinal monitoring dataset and a mature-stem harvesting

TABLE II
EVALUATION PLAN FOR THE INTEGRATED SYSTEM

Experiment	Comparison	Primary evidence
Modality ablation	Sensor vs. morphology vs. RGB vs. fusion	Macro-F1, AUROC, calibration
Plant hold-out	Seen vs. unseen plant IDs	Generalisation without leakage
Environment hold-out	Training environments vs. new environment	Domain robustness
Bark localisation	Detection/segmentation variants	IoU/Dice, target error
Robot trial	Automatic vs. supervised baseline	Peel success, damage, time, aborts

dataset rather than imply that all monitored young plants were followed through a complete commercial maturity cycle.

All observations from one plant or stem are kept in the same split:

$$P_{train} \cap P_{val} = P_{train} \cap P_{test} = P_{val} \cap P_{test} = \emptyset. \quad (8)$$

This plant-wise split prevents identity leakage. When multiple chambers or sites are available, a leave-one-environment-out experiment is added to test robustness to lighting, background, sensor calibration, and cultivation differences.

B. Evaluation metrics

Readiness classification is evaluated with precision, recall, macro-F1, balanced accuracy, AUROC, and probability calibration. Precision, recall, and F1 are

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (9)$$

$$F_1 = 2\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (10)$$

Continuous growth targets, if used, are reported with MAE, RMSE, and R^2. Bark-region detection is assessed using IoU/Dice, while camera-to-robot calibration is measured by Euclidean target error

$$e_{pos} = \|\mathbf{p}_{pred} - \mathbf{p}_{gt}\|_2. \quad (11)$$

Physical trials record peel initiation, target-length peel completion, bark/wood damage, cycle time, localisation failures, safety aborts, and operator overrides.

V. Prototype Implementation and Current Validation Status

The proof-of-concept is organised around a Python inference service, image-processing utilities, sensor acquisition, a web dashboard, and a robot-control interface. Sensor readings and images are linked by identifiers and timestamps so that the origin of each recommendation can be audited. The learning service returns a readiness decision together with model confidence; the vision service returns a bark region and geometric target; and the controller records whether an action was executed, rejected, aborted, or overridden.

This modular design is important because an incorrect final harvest can arise from very different causes. A plant may be classified incorrectly even when the stem is localised perfectly. A correct readiness decision may be followed by a poor camera-to-robot transform. Finally, an accurate target may still fail during contact because of bark adhesion, stem curvature, or end-effector mechanics. The trial log therefore assigns failures to readiness, visual localisation, calibration, motion, contact/peeling, or safety categories.

At the current stage, the project should be treated as a proof-of-concept rather than a completed plantation-scale validation. The software pipeline, model-serving interface, dashboard flow, and robot-guidance logic can be developed and demonstrated independently, but conference-quality numerical claims require a locked plant-wise test set, reproducible prediction logs, calibration measurements, and repeated physical trials. For this reason, example figures that appeared in early proposal documents should not be reported as measured results unless the corresponding evidence is available.

A. Statistical analysis and reproducibility

Model selection is performed only on training and validation plants. The final test partition remains locked until preprocessing, model architecture, hyperparameters, and decision thresholds are fixed. Where sample size permits, uncertainty is estimated using bootstrap resampling at the plant level, not by resampling correlated images from the same plant. Paired model comparisons use the same held-out plants.

The reproducibility package should preserve raw timestamped sensor files, original images, annotation files, plant/stem identifiers, exact data splits, calibration records, model versions, random seeds, saved test predictions, and robot-trial logs. Scripts used to calculate every reported metric should be retained together with the manuscript. This is particularly important for a system that combines several modules, because a dashboard screenshot alone cannot establish scientific performance.

VI. Implementation and Integration Details

A. Data exchange and traceability

The integrated system uses a small number of explicit messages rather than allowing each module to read directly from another module's internal variables. This reduces coupling between the IoT, ML, vision, and robot components. A sensor message contains the plant or stem identifier, environment identifier, timestamp, sensor values, and a quality flag. An image-analysis request carries the same biological identifier together with the image identifier and model version. The response contains the predicted health/readiness state, confidence, the detected bark region, and the calibration record used to obtain the physical target. Robot feedback then records the commanded state, execution result, duration, and any safety event. This message-level trace makes it possible to reconstruct why a particular action was recommended.

TABLE III
RECOMMENDED FIELDS RETAINED FOR EACH DECISION-TO-ACTION RECORD

Field group	Example retained information
Identity	plant/stem ID, environment ID, timestamp
Sensor context	temperature, RH, soil moisture, light, quality flags
ML output	model version, readiness probability, health class, thresholds
Vision output	image ID, bark region, confidence, calibration ID
Robot command	target pose, state, speed/limit profile
Trial outcome	completed/rejected/aborted, time, failure category, override

The dashboard is used as a supervision and audit interface rather than as evidence of model accuracy. It can display current environmental conditions, plant identity, growth/readiness estimates, bark-health output, the detected stem region, robot state, and recent warnings. When the operator reviews a case, both the visual output and the underlying numeric prediction are stored. This avoids a common prototype problem in which only a screenshot is retained while the model version, threshold, or sensor history that produced the screenshot is lost.

B. Robot state machine

The harvesting controller is represented as a state machine instead of a single movement command. In the `OBSERVE` state, a fresh image and the latest sensor context are acquired. `DECIDE` evaluates harvest readiness and health constraints. If the decision is acceptable, `LOCALISE` estimates the usable bark region and converts it to the robot frame. `VALIDATE` checks the target against the calibrated workspace and confirms that confidence remains above threshold. The robot then moves through `APPROACH`, `CONTACT`, and `PEEL`. Any abnormal force/current, lost target, timeout, or operator intervention moves the controller to `ABORT`, followed by a retreat to a known safe pose. This sequence makes safety conditions visible in the experimental log and prevents a classification result from being translated directly into uncontrolled motion.

The same state machine also supports controlled comparison with human-assisted operation. For example, an experiment can keep the vision and calibration stages fixed while a person confirms the target before `APPROACH`. A second experiment can allow automatic target approval but retain manual confirmation of readiness. These partial-automation baselines help identify whether errors originate from the biological decision, the geometric estimate, or the mechanical peeling interaction.

C. Failure taxonomy and diagnostic evaluation

An end-to-end success rate is useful, but it is not sufficient for improving the system. Each failed attempt is therefore assigned to one primary category: (i) *readiness error*, when a biologically unsuitable stem is accepted or a suitable stem is rejected; (ii) *health-classification error*, when a labelled bark condition is misclassified; (iii) *localisation error*, when the visible bark region is identified incorrectly; (iv) *calibration error*, when a correct image location is mapped to an inaccurate physical target; (v) *motion/control error*, when the robot cannot reach or maintain the commanded pose; (vi) *peeling interaction failure*, when contact is reached but the bark cannot be removed to the required length or quality; and (vii) *safety abort*. A failure matrix of this kind is more informative than combining all cases under a single "unsuccessful harvest" label.

For field-oriented testing, nuisance factors should be recorded rather than treated as unstructured noise. Relevant factors include illumination level, viewing angle, stem diameter, partial occlusion, background clutter, wet/dry bark condition, and camera-to-stem distance. Stratifying results by these factors can reveal whether a model that performs well overall fails systematically in particular operating conditions. It also provides a direct path for future improvement: additional training data address perception weaknesses, recalibration addresses geometric drift, and end-effector or control redesign addresses contact failures.

D. Research questions for the final evaluation

The final experiments are organised around four research questions. **RQ1** asks whether environmental history contributes predictive information beyond morphology and RGB appearance. **RQ2** asks whether multimodal fusion improves harvest-readiness generalisation on unseen plants and unseen environments. **RQ3** asks whether the calibrated vision pipeline can place robot targets on the intended bark region with sufficiently small physical error for controlled peeling. **RQ4** asks whether readiness and confidence gating reduce inappropriate or unsafe robot actions compared with a vision-only trigger. These questions keep the evaluation focused on the system's actual novelty rather than on isolated neural-network accuracy.

VII. DISCUSSION

The proposed architecture changes the role of machine learning from a stand-alone classifier to a decision layer inside a physical harvesting system. This has three consequences. First, false positives are more serious than in a passive monitoring application because an incorrect "ready" decision can trigger a physical action. Threshold selection should therefore consider the cost of unnecessary harvesting, not only overall accuracy. Second, multimodal fusion is useful only if it improves performance on genuinely unseen plants or environments. An apparent gain caused by repeated observations from the same plant would not demonstrate generalisation. Third, coordinate calibration must be evaluated separately from image recognition; a visually plausible line or point outside the actual stem is a geometric failure even if the classification label is correct.

The cinnamon domain also creates opportunities that differ from common fruit-harvesting systems. Harvest readiness depends on a history of growth and bark condition, while peeling is a contact-rich manipulation task. This makes a staged

architecture—monitor, decide, localise, validate, manipulate—more appropriate than a single end-to-end command from an image. It also allows future components to be replaced individually: a stronger sensor model, a better segmentation network, or a more capable end effector can be introduced without redesigning the complete pipeline.

VIII. Limitations and Ethical/Safety Considerations

The main limitation is biological time. Cinnamon is perennial, so a short project cannot reproduce every stage of commercial maturity under all agro-ecological conditions. Ground-truth readiness can also be subjective unless expert judgement is paired with a repeatable structural or peelability protocol. Models trained in controlled environments may degrade under field illumination, rain, wind, occlusion, disease, and variation in stem shape. In addition, bark peeling is mechanically different from fruit picking and requires specific contact-control validation.

The system does not collect sensitive human data, but physical safety is still a core research requirement. Robot trials should use a restricted workspace, emergency stop, conservative speed and force limits, supervised operation, and explicit manual override. Any autonomous harvesting claim should be limited to the conditions that were actually tested.

IX. Conclusion

This paper presents a unified research framework for AI-driven cinnamon growth monitoring and automated harvesting. The approach links IoT sensing, multimodal growth and harvest-readiness modelling, bark-health analysis, calibrated vision, and a safety-gated robotic controller. Its main novelty is not any single sensor or neural network; it is the traceable decision-to-action path that requires biological readiness and geometric confidence before the robot is permitted to peel. A rigorous evaluation therefore has to combine plant-wise predictive testing, environment hold-out experiments, bark localisation, physical coordinate error, and repeated robot trials. Completing those measurements with reproducible logs will determine whether the system can move from a laboratory proof-of-concept toward reliable field use in cinnamon cultivation.

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